Kaifeng Jia

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Education

M.Eng. in Electrical and Computer Engineering
Cornell University — *Aug 2025 – Dec 2026 (Expected)*

- Selected coursework: Fast Robots, Autonomous Mobile Robots, Embedded Operating Systems, Natural Language Processing.

B.Eng. in Optoelectronic Information Science and Engineering
Hefei University of Technology — *Sep 2021 – May 2025*

- Selected coursework: Signals and Systems, Digital Image Processing, Sensor Principles and Technology.

Technical Skills

Robot Learning: Deep learning, imitation learning, reinforcement learning, multi-agent RL, PPO
Robotics: SLAM, state estimation (EKF, Bayes filters), sensor fusion (IMU, ToF), planning, PID control
Programming: Python, C/C++, Swift, MATLAB
Frameworks & Tools: PyTorch, Isaac Lab / Isaac Sim, ARKit, Git, Linux

Research Experience

Cooperative Multi-Agent Humanoid Manipulation — Carnegie Mellon University
Research Mentor: Hao E. Zhang — *Jun 2026 – Present*

- Retargeted OMOMO human-object interaction demonstrations to Unitree G1 humanoids using OmniRetarget and trained single-agent whole-body motion-tracking policies for push and pull tasks in Isaac Sim.
- Initialized a multi-agent reinforcement learning framework from pretrained imitation-learning checkpoints to investigate emergent cooperative behaviors without explicitly assigned roles.

Physics-Aware Architecture for Metalens-Based Monocular Depth — New York University
Research Mentor: Bingxuan Li — *May 2026 – Present*

- Built on *Physically Grounded Monocular Depth via Nanophotonic Wavefront Encoding* to explore PromptDA-inspired architectural adaptations for polarization-encoded optical cues.
- Designed a physics-aware optical head that estimates x- and y-direction shifts through symmetric local correlation and fuses confidence-aware features into the DPT decoder for metric depth estimation.

Projects

Fast Robots: Autonomous RC Car Platform — Cornell University, Jan 2026 – May 2026

- Built an end-to-end autonomy stack with ToF-based 360° environment mapping, Bayes filter localization on a real robot, and navigation through nine waypoints using orientation PID and an onboard state machine.
- Designed distance and orientation controllers with a Kalman filter using system-identified drag and momentum parameters; integrated ToF and IMU sensors, motor drivers, and BLE on an Artemis microcontroller.

Indoor AR Navigation System for Olin Library — Cornell University, Sep 2025 – May 2026

- Built an iOS indoor navigation system using ARKit visual-inertial SLAM to estimate device pose, support map-based relocalization, and render real-time guidance on the live camera feed.
- Constructed a metric-scale walkable graph from floor plans and applied Dijkstra's algorithm for path planning, turn detection, and 3D AR directional-cue rendering.

Smart Desk Assistant: Multimodal Raspberry Pi System — Cornell University, Aug 2025 – Dec 2025

- Built a Raspberry Pi desktop assistant integrating a PyGame/PiTFT touch UI, Vosk speech recognition, an LLM API, text-to-speech, and PID-controlled DNN face tracking.